MAE386
Ideation, Thought Media, and Mathematics

Final Examination

April 2026

Instructions

1. This is a 3-hour Final Examination. There are 16 questions and 1 bonus question. Answer the first 16 questions neatly on this examination question sheet. You can give the bonus question a go if you have the time and inclination. Note that, for the bonus question, part (a) must be answered correctly to get marks for part (b).
2. You must provide reasoning for your answers. Correct final answers, without reasoning, will receive no marks.

Question	Out of	Mark
1	4	
2	2	
3	4	
4	5	
5	8	
6	4	
7	3	
8	3	
9	2	
10	6	
11	6	
12	4	
13	4	
14	2	
15	4	
16	4	
Bonus	5	
Total	65	

Question 1 (4 marks)
We suggested that exographics has two values in the discovery of certain kinds of ideas. What are those values called and how are they defined?

Question 2 (2 marks)
We use the word "persistent" in our definition of exographics. What does persistent mean in this context? In your explanation, give an example of text that is not persistent.

Question 3 (4 marks)
Consider this argument.

> We can make arguments aurally, but speech has the cognitive property that it disappears almost as quickly as it's spoken. Hence, it is very difficult to make arguments in our aural field.

Explain why this argument is incorrect using the example of solving the linear equation

$$2x - 6 + 4x = 12 + 10x - 14 - 8x + 4$$

with two sound-recording devices, each with a playback feature.

Question 4 (5 marks)
Define the following terms:

(a) cultural object

(b) made world

(c) lifeways

(d) Ideasphere

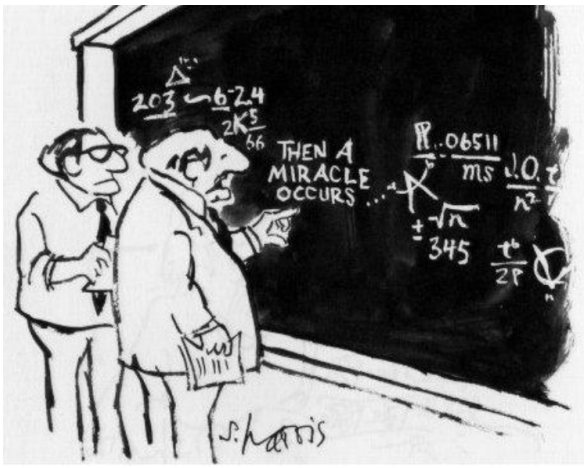
(e) the e-Class

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Question 5 (8 marks)
Assume that each cultural object described below is in the Ideasphere. For each, assess whether it belongs to the e-Class and give your reasoning why or why not.

(a) any one of the models that implement OpenAI's ChatGPT

(b) Sidney Harris's iconic math cartoon:



"I think you should be more explicit here in step two."

(c) the nursery rhyme *The Cat and the Fiddle*

It begins "Hey diddle diddle, the cat and the fiddle, ..." The earliest evidence of it in print dates to 1765. The rhyme fits into a long tradition of nonsense rhymes originating in Tudor and Stuart England (1500s–1600s). Apparently, many 16th and 17th century English taverns were called The Cat and Fiddle.

(d) eating bone marrow

Very early in our species's run and likely earlier, our ancestors realized that bone marrow was filling and nutritious. We know today that it's calorie dense in fats and important minerals. In prehistoric times, marrow was also storeable if the bones it came in were placed in cool places. Today marrow is considered a delicacy in some high-end restuarants.

(e) The TWINSCAN EXE:5200 (High-NA EUV)

This machine, built and designed by the Dutch firm ASML, does lithography on microchips. It costs $380 million, weighs 165 tons, and takes 250 engineers about 6 months to assemble.

(f) the drop spindle

Drop spindles are used to make yarn from various kinds of fibers including sheep wool and plant-based fibers. There is a fascinating material science associated with spinning such fibers. When they are intertwined with spinning, they end up forming much stronger, continuous yarn. A modern drop spindle looks like this: There is archaeological evidence that drop spindles



were used at Nahal Ein-Gev II in the Levant about 12,500 years ago. There is also evidence of twisting fibers long before the appearance of drop spindles. This evidence dates to about 30,000 years ago in the Caucasus where flax fibers were spun into linen.

(g) Berthillon ice-cream

You can only get Berthillon ice-cream on the Île Saint-Louis in Paris. Their Wild Strawberry (Frais des bois) is to die for. Raymond Berthillon began making his ice-cream in the back of his store in 1954. There is evidence that he was very systematic. He wrote down recipes adjusting them for fat content, and sugar levels. He was very careful about temperature curves (good ice-cream requires that you vary its temperature as you make it) and was continually refining texture and overrun (overrun is the incorporation of air into an ice-cream as you make it and is known to affect how fat crystals and ice crystals interact on the tongue.) He used notebooks continually to record what worked and what didn't.

(h) the epithet "Float like a butterfly, sting like a bee."

It's usually attributed to Muhammad Ali's cornerman, Drew "Bundini" Brown and it first appeared in 1964 before Ali (then Cassius Clay) fought Sonny Liston for the first time. It's a poetic reference to Ali's boxing style which combined movement and quick, punishing flurries of punches.

Question 6 (4 marks)
In the course, we discussed Daniel Kahneman's dual purpose theory of thinking. Explain what he meant by System 1 and System 2 thinking and give an example of each.

Question 7 (3 marks)
Define the dialectic and diagraphic and explain how they are useful to us in the discovery of ideas.

Question 8 (3 marks)
What is a phonetic writing system and what is its main advantage over a pictographic system?

Question 9 (2 marks)
One of the fortunate outcomes of human evolution is that our physiological development made it possible for us to make about 40 different sounds which we could string together to form words and communicate. In fact, we communicate in an extraordinarily refined way relative to all other species. Had we not evolved the ability to make the different sounds, would we be communicating as effectively as we do today?

Question 10 (6 marks)

(a) Suppose you are a captain in the Canadian Armed Forces and one of your lieutenants writes these two sentences at different points in an email he sent to you:

Sentence 1: "The reason why he was late is that he overslept."

Sentence 2: "The two of them collaborated together on the project."

Even though the meanings of these sentences are clear, you decide to explain to him how to write them with fewer words. But before doing so, you decide to rewrite them yourself. How would you rewrite them with fewer words?

(b) Rewrite the following sentence correctly with fewer words:

"His toolkit is comprised of the habits, heuristics, and external scaffolds he builds in order to think."

Question 11 (6 marks)
Is there an argument to be made for a global techno-literate culture? In other words, is there a sense in which the nations of the developed world form a single techno-literate culture? Begin your answer by describing the characteristics of a techno-literate culture.

Question 12 (4 marks)

In *Catching Unicorns*, we argue a theory of embodied ideation, a process that requires the mind, the body, and artifacts in the environment (i.e. exographics). Ideation, exographics, and literacy definitely play a role in the work of Canadian Armed Forces officers. Assess whether the Canadian Armed Forces is a techno-literate culture.

Question 13 (4 marks)

There were a number of wonderful scholars at the University of Toronto over the mid to late 20th century. They included Harold Innis, Eric Havelock, and Marshall McLuhan among others. Early in his career at St. Michael's College, McLuhan met Innis and read his *Empire and Communications* (1950) wherein Innis argues that different media technologies, like papyrus, parchment, vellum, etc., influenced the organization and power structures of empires. This set McLuhan to think about the role of media in society and he eventually published his ideas in *Understanding Media: The Extensions of Man* (1964). These scholars, known collectively as "the Toronto School," were ahead of their time, realizing that media were important to thought and to the organization of society and progress of culture. Based on what we've covered in the course, would you say they were correct? And if so, why?

Question 14 (2 marks)

At various points in the course, we've looked at societies which are not able to produce e-Class cultural objects. Does this mean that these people are limited cognitively? Take the position that they are not and give two arguments to support that conclusion.

Question 15 (4 marks)

We've talked a lot about human imagination over the course. It's this marvelous ability we have to bring phantasms of our mind's making to the surface. Dickens imagined Ebeneezer Scrooge, the main character in *A Christmas Carole*. Newton imagined the Law of Universal Gravitation, his idea that a force of attraction existed between any two objects, one he called "gravitation." It's why an apple falls to the floor when we drop one. It's why the Moon revolves around the Earth and the Earth around the Sun. Apparently it also plays a role in tides. If you were asked what keeps the moon in orbit, you would, no doubt, answer "gravitation!" But does that really explain how the moon stays in orbit? To me it doesn't. Nobody has seen gravitation. Nobody has offered a deeper explanation of how it might work to keep the moon in orbit. There have been other explanations including Einstein's based on the curvature of space-time. But they also suffer the same weakness. Still another explanation is that it's God's invisible hand at work. That is, if we can't explain a phenomenon "scientifically," we resort to God and/or magic. Over our intellectual history, we've gradually supplanted God and magic with "science," yet the science is not very satisfactory as I've suggested above. Why then do we do science? Write at most a paragraph explaining what my argument misses. You might consider beginning your answer with this sentence: "Science doesn't have to fully explain a phenomenon to be useful."

Question 16 (5 marks)

The narrative of *Catching Unicorns* is in two parts. In the first, we present a theory of ideation that centers on exographics. In the second, we argue the connection between our ideation ability with exographics and the rise of techno-literate culture. In so doing, we hope to establish the importance of exographics as a crucially important tool of civilization, an idea that is generally frowned on by a significant number of scholars.

Military officers have a wonderful skill of taking a blank piece of paper and drawing a diagram to illustrate relationships among concepts. I would like you to create such a diagram to explain the first part of our narrative, that exographics enables us to "catch unicorns" we otherwise couldn't. You might consider including some of these concepts: individual mind, network of minds, ideas, ideation, exographics, working memory, long-term memory, Ideasphere, e-Class ideas, the diaconatic, the diagraphic, the dialectic, social intelligence, and general intelligence. I have no idea how I would do this but I'm hoping you do.

Bonus Question (5 marks)
[Note: to get any marks on this question, you must answer part (a) correctly.]
Consider a triangle and its opposite side bisectors (from each vertex a line is drawn that divides the opposite side into lines of equal length). Ancient Greek mathematicians noticed that these lines intersected at a single point. More generally, they invented a system of analysis with definitions (i.e., points, lines, triangles, and so on), axioms (i.e., parallel lines do not intersect), and then theorems (the opposite side bisectors of any triangle intersect at one point). This is the same system we use today in mathematics.

(a) Suppose the vertices of a triangle are at (6,6), (12,6), and (6,18). An opposite-side bisector is called a median. Let the centroid be given by the average of the three vertices: ([6+12+6]/3, [6+6+18]/3) = (8,10). Can you show that each of the three medians goes through this centroid?

(b) Once you have shown this, assess whether you could have done it without pencil and paper.